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KAHRAMAA

المؤسسة العامة القطرية للكهرباء والماء

Qatar General Electricity & Water Corporation

QATAR POWER TRANSMISSION SYSTEM EXPANSION

POWER EVACUATION FROM DUKHAN SOLAR PROJECT- PACKAGE C1

CONTRACT No. GTC/1266D/2025

CONSULTANT:

ENERGOPROJEKT ENTEL LTD

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Factory Acceptance Test Procedures for Power Cable System (Cable / Accessories)

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1. Factory Acceptance Test procedure

1.1. 220kV XLPE Power Cable

1.1.1. Routine test

NO	Test Items	Reference Standard	Equipment	Acceptance Criteria	Procedure
1)	Voltage test	IEC 62067: 2022 C 9.3	AC 750kV Voltage tester	No breakdown at 325kV for 30min	a. Test shall be done at ambient temperature & power frequency (50Hz). b. Test voltage shall be raised gradually to the specified value(325kV) between the conductor and metallic sheath, and then held for 30min.
2)	Partial discharge test	IEC 62067: 2022 C 9.2, IEC 60885-3	AC 750kV Voltage tester PD Calibrator	Max.3pC at 195kV	a. Test shall be done before and after AC voltage test at ambient temperature using an alternating test voltage with power frequency (50Hz). b. Sensitivity shall be 3pC or better. Test voltage shall be raised gradually to and held at 227.5kV for 10 sec, then slowly reduced to 195kV
3)	Dielectric loss angle	IEC 62067: 2022 C 12.4.5	Tan delta tester	Max. 0.0005 at 195kV (1.5U0)	Dielectric loss angle shall be measured at ambient temperature and specified voltage(1.5U0)
4)	Conductor examination · Check of no. of wires · Ratio of diameters of two different wires · Conductor diameter	IEC 62067:2022 C 10.4, IEC 60228	Thickness meter Vernier calipers	No for wires: 91 Ratio: Max 2 Diameter: Nom 63.0mm	a. One sample shall be taken from a dispatch drum and dissected to the conductor. b. Conductor construction shall be checked.
5)	Measurement of electrical resistance of conductor	IEC 62067:2022 C 10.5, IEC 60228	Digital Low Resistance Ohmmeter	Max 0.0072Ω/km at 20℃	The d.c. resistance of the conductor shall be corrected to a temperature of 20 ℃ and a length of 1 km in accordance with the formulae and factors given in IEC 60228.
6)	Measurement of electrical resistance of metallic sheath	IEC 62067:2022 C 10.5, IEC 60228	Digital Low Resistance Ohmmeter	Max 0.06970 Ω/km at 20℃	The d.c. resistance of the metallic sheath shall be corrected to a temperature of 20 ℃ and a length of 1 km in accordance with the formulae and factors given in IEC 60228.
7)	Oversheath voltage withstand test	IEC 62067: 2022 C 9.4	DC Voltage tester	No breakdown at DC 25kV for 1min	DC Voltage of 25kV shall be applied for 1 min. between metallic layer and the graphite coating.

1.1.2 Additional Regular test

NO	Test Items	Reference Standard	Equipment	Acceptance Criteria	Procedure
1)	Measurement of thickness of insulation, oversheath & metallic sheath	IEC 60811-201 IEC 62067: 2022 C 10.6	Thickness meter Projector	Insulation Nom: 22.0mm Min: 19.8mm Metallic sheath Nom: 2.6mm Min: 2.47mm Oversheath Nom: 4.5mm Min: 3.73mm	a. The sample shall be taken from each end of every drum. b. Take a wafer from the insulation perpendicular to the conductor axis. c. The insulation thickness shall be measured with profile projector.
2)	Measurement of insulation concentricity	IEC 62067: 2022 C 10.6	Profile projector	Max 10%	a. The sample shall be taken from each end of every drum. b. 50mm sample shall be taken from the cable core. The sample shall be cut into wafers approximately 12.5mm thick and viewed under magnification. c. The samples shall be examined at six points equally spaced around the cable with one of the points corresponding to point of minimum insulation thickness, tmin. d. The maximum allowable deviation between the thickness at the point of minimum thickness, tmin, and the maximum measured thickness value, tmax, shall be less than 10% of the measured average thickness.
3)	Measurement of insulation purity	KM Spec. V3 S3.3.12.2 C1.4.4.8.5	Microscope	Void : Max. 0.02mm Number of Voids in Any Area of 10mm×10mm : Max. 10 Black and Metallic Particle : Max. 0.08mm	a. A sample approximately 50mm long shall be taken from the cable core. b. The sample shall be cut into 25 wafers approx. 2mm thick. c. The entire area of each wafer shall be examined with microscope.
4)	Measurement of moisture content in extruded insulation and screens	KM Spec. V3 S3.3.12.2 C1.4.4.8.6	Moisture tester	Insulation :Max. 150ppm Screens : Max. 500ppm	a. The sample shall be taken from sample drum. b. The moisture content of the insulation and screens shall be measured using the Karl Fischer titration method. Three measurements shall be taken from each of the two screens and insulation.
5)	Hot set test for XLPE insulation	IEC 60811-507 IEC 62067: 2022 C 10.9	Aging Oven Recorder	Elongation under Load: Max. 175 %, Permanent Elongation after Cooling: Max. 15 %	a. Dumb-bell test pieces shall be prepared from the insulation. b. The test pieces shall be suspended in the oven and weights shall be attached to the bottom of test pieces to exert the specified force. (20N/cm2) c. After 15min. in the oven at the temperature specified for the material, the distance between the marker lines shall be measured and the percentage elongation calculated. (Temperature: 200℃) d. The tensile force shall be removed from the test piece, and the test piece left to recover for 5 min. at the specified temperature. The test piece shall be removed from the oven and allowed to cool slowly to ambient temperature, after which the distance between the marker lines shall be measured again.

NO	Test Items	Reference Standard	Equipment	Acceptance Criteria	Procedure
6)	Insulation shrinkage test	IEC 60840:2020 C 12.5.16	Oven	Max. 4.5 %	a. The sample shall be taken from sample drum. b. All layers (outside of outer semi-conducting layer) shall be removed. c. Marks 200mm apart from each other shall be put and sample shall be put into an oven. d. Ageing condition: at 130±3℃ for 6 hours
7)	Measurement of resistivity of semiconducting screens	IEC 62067: 2022 C 12.4.11	Oven, Resistivity tester	Conductor Screen : Max. 1,000 Ω.m Insulation Screen : Max. 500 Ω.m	a. The sample shall be taken from sample drum. b. The sample is subjected to the ageing treatment. (100±2℃,168hrs) c. Electrodes shall be assembled according to IEC 62067. d. Sample shall be put into the oven and keep 30 min. with temperature 90℃, then measured.
8)	Measurement of screen protrusions	KM Spec. V3 S3.3.12.2 C1.4.4.8.10	Microscope	Shall be cylindrical, smooth and free from protrusions and irregularities which extend more than 0.03mm into the insulation	a. The approximately 50mm long sample shall be taken from the cable core. b. The sample shall be cut into 25 wafers approx. 2mm thick. c. The semi-conducting conductor and insulation screens of wafer shall be examined with microscope. d. 0.5m sample of outer (insulation) screen shall be taken from sample drum and this sample shall be rendered transparent by immersion in hot oil. e. The screen shall be examined visually.
9)	Measurement of capacitance	IEC 62067: 2022 C 10.10	Capacitance Meter	Max 0.2927㎌/km	a. Capacitance shall be measured between conductor and metallic sheath with a capacitance meter. b. The measured capacitance shall be converted to 1km length.
10)	Measurement of crosslinking by product concentration in XLPE cables	KM Spec. V3 S3.3.12.2 C1.4.4.8.12	TGA Analyzer	a. Total weight change of test sample after 30min : Max. 1.25% of the original sample weight b. Rate of change in the first 5min : Max.0.14 % c. Average rate of change of sample weight between the Time of 15 – 30min.: Max. 0.01%	a. One sample shall be taken from sample drum. b. The test is to be conducted using TGA apparatus. c. A 20mg insulation sample is required. The test is to be conducted for 30 min. at 150℃. d. The % loss of weight of the sample is to be recorded. e. Five samples are to be taken from radius direction, the result being the average of five tests.
11)	FTIR	KM Spec. V3 S3.3.12.2 C1.4.4.8.13	FTIR Analyzer	Comply with the Results of Type Test.	a. The sample shall be taken from both ends of every dispatch drum. b. The measurement shall be carried out using the FTIR equipment.
12)	DSC	KM Spec. V3 S3.3.12.2 C1.4.4.8.13	DSC Analyzer	Comply with the Results of Type Test.	a. The sample shall be taken from both ends of every dispatch drum.b. The measurement shall be carried out using the DSC equipment.

NO	Test Items	Reference Standard	Equipment	Acceptance Criteria	Procedure
13)	TGA	KM Spec. V3 S3.3.12.2 C1.4.4.8.1 3	TGA Analyzer	Comply with the Results of Type Test.	a. The sample shall be taken from every batch(batch means one extrusion run length or maximum 10drums) b. The measurement shall be carried out using the TGA equipment
14)	Cleanliness measurement on the incoming insulation material	KM Spec. V3 S3.3.12.2 C1.4.4.8.1 3	-	90 ~ 110μm / 100g: Max. 2 EA Bigger than 100μm / 100g: Max. 1 EA	One sample shall be taken from a batch of incoming material
15)	Smoothness measurement on the incoming semi-conducting material	KM Spec. V3 S3.3.12.2 C1.4.4.8.1 3	-	0.05 ~ 0.10mm / 200cm2 : Max. 10 EA 0.10 ~ 0.25mm / 200cm2 : Max. 5 EA Bigger than 0.26mm /200 cm2 : 0 EA	One sample shall be taken from a batch of incoming material
16)	Measurement of weight for copper and lead of cable construction	KM Spec. V3 S3.3.12.2 C1.4.4.8	ELECTRIC BALANCE	Copper: Approx 24.6kg/m Lead: Approx 12.0kg/m	a. The sample shall be taken from cable core. b. The weight shall be measured with a balance. c. The unit of a measured value shall be transformed to kg/m
17)	Measurement of density of HDPE & carbon content	KM Spec. V3 S3.3.12.2 C1.4.4.8	Density meter	Density : Min 0.94 g/cm³ Carbon content : 2.5±0.5%	a. One sample shall be taken from sample drum. b. The density is measured using a density meter.
18)	Water Penetration Test	IEC 62067: 2022 C 10.13, Annex E & Annex F	Recorder	No water emerged cable ends	Annex E a. One sample shall be taken per 10km order. b. Cable sample shall be prepared according to IEC 62067. c. 10 heating cycles shall be applied to the cable sample after 24 hours stand without heating cycle d. 1 Cycle: 8 hours heating + 16 hours natural cooling Annex F a. The tube shall be filled within 5 min with tap water at a temperature of (20 ± 10) °C so that the height of the water in the tube is (1,0 ± 0,050) m above the cable centre b. The sample shall be allowed to stand for 11 days at ambient temperature.

1.1.3 Sample test

NO	Test Items	Reference Standard	Equipment	Acceptance Criteria	Procedure
1)	Power frequency voltage test	KM Spec. V3 S3.3.12.2 C1.4.4.8.1 4	Voltage tester	No breakdown at 650kV(5U0) for 1hour	a. Test shall be done at ambient temperature. b. Test voltage of 650kV shall be applied between conductor and metallic sheath on 20m sample.

※ Samples are to be selected from completed production immediately before test; manufacturer pre-selection is in general not allowed.

1.2. Accessories for 220kV XLPE Power Cable

1.2.1. Routine test(Joint)

NO	Test Items	Reference Standard	Equipment	Acceptance Criteria	Procedure
1)	FTIR	KM Spec. V3 S3.3.12.2 C1.4.5.1	FTIR Analyzer	Comply with the Results of Type Tests	a. The tests shall be carried out on one joint in every ten manufactured on incoming material. b. Test piece shall be prepared and measured for FTIR.
2)	Cleanliness measurement on the incoming insulation material	KM Spec. V3 S3.3.12.2 C1.4.5.1	-	90 ~ 110μm: Max. 2 EA Bigger than 100μm: Max. 1 EA	One sample shall be taken from a batch of incoming material.
3)	DLA measurement of insulation at 90ºC	KM Spec. V3 S3.3.12.2 C1.4.5.1	DLA Analyzer	Comply with the Results of Type Tests	a. The tests shall be carried out on one joint in every ten manufactured on incoming material. b. Test piece shall be prepared and measured DLA
4)	TGA	KM Spec. V3 S3.3.12.2 C1.4.5.1	TGA Analyzer	Comply with the Results of Type Tests	a. The tests shall be carried out on one joint in every ten manufactured on incoming material. b. Test piece shall be prepared and measured TGA

1.2.2. Additional test(Joint)

NO	Test Items	Reference Standard	Equipment	Acceptance Criteria	Procedure
1)	HV withstand test at 2.5U0 for 30 minutes	KM Spec. V3 S3.3.12.2 C1.4.5.1.1	AC 600kV Voltage tester	No breakdown at 325kV for 30min	a. One piece joint shall be installed on the cable mounted to test termination in advance. b. Specified test voltage(325kV) shall be applied to the conductor for 30min.
2)	PD measurement at 1.5 U0	KM Spec. V3 S3.3.12.2 C1.4.5.1.1	AC 600kV Voltage tester PD Calibrator	Max. 5pC at 195kV	a. One piece joint shall be installed on the cable mounted to test termination in advance. b. Partial discharge level shall be measured at 195kV.
3)	X-Ray test	KM Spec. V3 S3.3.12.2 C1.4.5.1.1	X-Ray	Any Crack or Crack Development Shall Not Be Found.	One piece joint shall be X-Rayed.
4)	Mechanical stretch of each rubber component	KM Spec. V3 S3.3.12.2 C1.4.5.1.1	-	No Damage	One piece joint shall be installed on the cable mounted to test termination in advance.
5)	Resistance of outer semi-conducting screen	KM Spec. V3 S3.3.12.2 C1.4.5.1.1	ohm meter	Max. 20kΩ.	The resistance of semi-conducting screen shall be measured with ohm meter.
6)	Integrity of insulated gap 25kV DC for 15 minutes	KM Spec. V3 S3.3.12.2 C1.4.5.1.1	DC voltage tester	No breakdown occurs at D.C. 25kV for 15min	Specified DC voltage shall be applied between gaps for 15 minutes.
7)	Packing and Preservation Inspection	KM Spec. V3 S3.3.12.2 C1.5	-	The quantities of the products shall be satisfied with packing list and Preservation requirements	a. The quantities of the products shall be satisfied with packing list. b. Type/Weight/Manufacturing contract information shall be provided by approved format(Shipping mark, Packing list)QC verification activity for preservation and storage

1.2.3. Routine test(Termination)

NO	Test Items	Reference Standard	Equipment	Acceptance Criteria	Procedure
1)	FTIR	KM Spec. V3 S3.3.12.2 C1.4.5.1	FTIR Analyzer	Comply with the Results of Type Tests	a. The tests shall be carried out on one sealing end in every ten manufactured on incoming material. b. Test piece shall be prepared and measured for FTIR.
2)	Cleanliness measurement on the incoming insulation material	KM Spec. V3 S3.3.12.2 C1.4.5.1	-	90 ~ 110μm: Max. 2 EA Bigger than 100μm: Max. 1 EA	One sample shall be taken from a batch of incoming material.
3)	DLA measurement of insulation at 90°C	KM Spec. V3 S3.3.12.2 C1.4.5.1	DLA Analyzer	Comply with the Results of Type Tests	a. The tests shall be carried out on one sealing end in every ten manufactured on incoming material. b. Test piece shall be prepared and measured DLA
4)	TGA	KM Spec. V3 S3.3.12.2 C1.4.5.1	TGA Analyzer	Comply with the Results of Type Tests	a. The tests shall be carried out on one sealing end in every ten manufactured on incoming material. b. Test piece shall be prepared and measured TGA

1.2.4. Addtional test(Termination)

NO	Test Items	Reference Standard	Equipment	Acceptance Criteria	Procedure
1)	HV withstand test at 2.5U0 for 30 minutes	KM Spec. V3 S3.3.12.2 C1.4.5.1.1	AC 600kV Voltage tester	No breakdown at 325kV for 30min	a. One piece sealing end shall be installed on the cable mounted to test termination in advance. b. Specified test voltage(325kV) shall be applied to the conductor for 30min.
2)	PD measurement at 1.5 U0	KM Spec. V3 S3.3.12.2 C1.4.5.1.1	AC 600kV Voltage tester PD Calibrator	Max. 5pC at 195kV	a. One piece sealing end shall be installed on the cable mounted to test termination in advance. b. Partial discharge level shall be measured at 195kV.
3)	X-Ray test	KM Spec. V3 S3.3.12.2 C1.4.5.1.1	X-Ray	Any Crack or Crack Development Shall Not Be Found.	One piece sealing end shall be X-Rayed.
4)	Packing and Preservation Inspection	KM Spec. V3 S3.3.12.2 C1.5	-	The quantities of the products shall be satisfied with packing list and Preservation requirements	a. The quantities of the products shall be satisfied with packing list. b. Type/Weight/Manufacturing contract information shall be provided by approved format(Shipping mark, Packing list)QC verification activity for preservation and storage

1.2.5. Routine test(Link Box)

NO	Test Items	Reference Standard	Equipment	Acceptance Criteria	Procedure
1)	Visual and dimensional inspection	KM Spec. V3 S3.3.12.6 C1.9.2.a	Measuring tape	Shall be meet with approved drawings	a. The dimensions of link box shall be measured with micrometer. Main measurements should be checked depending on the approved drawings. Measurement deviations have to be within the given limits. b. Thickness of painting should be greater than 80 micron.
2)	Insulation Resistance measurement test	KM Spec. V3 S3.3.12.6 C1.9.2 b	Insulation resistance measurement	Min. 10MΩ	a. DC 25kV voltage shall be applied for 5min. directly between each combination of links and between the links and earth without bonding reads. b. If SVL include the test object, it is dismounted in this test.
3)	Contact resistance measurement test	KM Spec. V3 S3.3.12.6 C1.9.2 c	Contact resistance tester	Max. 10μΩ	a. DC 50A Current will be applied to each contact points which have been squeezed with 40kN torque. b. If exist SVL shall be dismounted during the test. c. The contact resistance shall be measured.
4)	Water sealing test	KM Spec. V3 S3.3.12.6 C1.9.2 d	Water tank	No leakage of the air from the sample during the test.	a. The steel cabinet shall be closed with its cover and seal by cap in each hole for bonding cable. b. The hole for earthing cable shall be connected to the air pressure device. c. The sample shall be taken in the water. d. The sample shall be maintained in 0.3bar(0.31kgf/cm2) pressure for 15min.

1.2.6. Additional test(Link Box)

NO	Test Items	Reference Standard	Equipment	Acceptance Criteria	Procedure
1)	The Impulse Voltage Withstand Test	KM Spec. V3 S3.3.12.6 C1.9.3	Impulse tester	No flash over - Between links: ± 40kVp / 10 times. - Between links and earth: ± 20kVp / 10 times	a. Specified impulse voltage should be applied directly between links and earthing terminal including a frame of the Link box without bonding reads. b. Test condition - Between links: ± 40kVp / 10 times. - Between links and earth: ± 20kVp / 10 times
2)	The DC Voltage Withstand Test	KM Spec. V3 S3.3.12.6 C1.9.3	DC Voltage tester	No flash over	a. DC 25kV voltage shall be applied for 5min. directly between each combination of links and between the links and earth without bonding reads. b. If SVL include the test object, it is dismounted in this test.
3)	Packing and Preservation Inspection	KM Spec. V3 S3.3.12.6 C1.10	-	The quantities of the products shall be satisfied with packing list and Preservation requirements	a. The quantities of the products shall be satisfied with packing list. b. Type/Weight/Manufacturing contract information shall be provided by approved format(Shipping mark, Packing list)QC verification activity for preservation and storage

1.2.7 Additional test(Sheath Voltage Limiters)

NO	Test Items	Reference Standard	Equipment	Acceptance Criteria	Procedure
1)	Leakage Current Test at DC 5kV for 1 min.	KM Spec. V3 S3.3.2	DC Voltage Tester	Max. 0.1mA	a. Specified DC voltage shall be applied between either sides of SVL for 1 minute. b. Leakage current shall be measured at the end of the test.
2)	Insulation Resistance	KM Spec. V3 S3.3.2	Insulation resistance measurement	Min. 10MΩ	a. Specified DC 1kV voltage shall be applied between either ends of SVL. b. Insulation resistance shall be measured at the end of the test.
3)	DC Voltage test	Comment	Voltage tester	Leakage current shall be less than 1mA at specified DC voltage.	a. Specified DC voltage shall be applied across the SVL. b. DC voltage shall be raised continuously while leakage current does not reach 1mA

Inspection and Test Plan

2. Inspection and Test Plan

2.1. Test Plan for 220kV XLPE Power cable and accessories

2.1.1. Routine Test (220kV Power cable)

Table 2-1 Routine Test

NO	Test Items	Reference Standard /Document	Equipment	Acceptance Criteria	Sampling	Inspection Involvement	
						LS	KM
1)	Voltage test	IEC 62067: 2022 C 9.3	AC 750kV Voltage tester	No breakdown at 325kV for 30min	LS: 100% FAT: 10% of Shipping drums	H	(W)
2)	Partial discharge test	IEC 62067: 2022 C 9.2	AC 750kV Voltage tester PD Calibrator	Max.3pC at 195kV		H	(W)
3)	Dielectric loss angle	IEC 62067: 2022 C 12.4.5	Tan delta tester	Max. 0.0005 at 195kV (1.5U0)		H	(W)
4)	Conductor examination · Check of no. of wires · Ratio of diameters of two different wires · Conductor diameter	IEC 62067:2022 C 10.4, IEC 60228	Thickness meter Vernier calipers	No for wires: 91 Ratio: Max 2 Diameter: Nom 63.0mm		H	(W)
5)	Measurement of electrical resistance of conductor	IEC 62067:2022 C 10.5, IEC 60228	Digital Low Resistance Ohmmeter	Max 0.0072Ω/km at 20℃		H	(W)
6)	Measurement of electrical resistance of metallic sheath	IEC 62067:2022 C 10.5, IEC 60228	Digital Low Resistance Ohmmeter	Max 0.06970Ω/km at 20℃		H	(W)
7)	Oversheath voltage withstand test	IEC 62067: 2022 C 9.4	DC Voltage tester	No breakdown at DC 25kV for 1min		H	(W)

2.1.2. Additional Regular Test (220kV Power cable)

Table 2-2 Additional Regular Test

NO	Test Items	Reference Standard /Document	Equipment	Acceptance Criteria	Sampling	Inspection Involvement	
						LS	KM
1)	Measurement of thickness of insulation, oversheath & metallic sheath	IEC 60811-201 IEC 62067: 2022 C 10.6	Thickness meter Projector	Insulation Nom: 22.0mm Min: 19.8mm Metallic sheath Nom: 2.6mm Min: 2.47mm Oversheath Nom: 4.5mm Min: 3.73mm	LS: 100% FAT: 10% of Shipping drums	H	(W)
2)	Measurement of insulation concentricity	IEC 62067: 2022 C 10.6	Profile projector	Max 10%	LS: 100% FAT: 10% of Shipping drums	H	(W)
3)	Measurement of insulation purity	KM Spec. V3 S3.3.12.2 C1.4.4.8.5	Microscope	Void : Max. 0.02mm Number of Voids in Any Area of 10mm×10mm : Max. 10 Black and Metallic Particle : Max. 0.08mm	LS: 20% of every extrusion run FAT: 10% of Shipping drums	H	(W)
4)	Measurement of moisture content in extruded insulation and screens	KM Spec. V3 S3.3.12.2 C1.4.4.8.6	Moisture tester	Insulation :Max. 150ppm Screens : Max. 500ppm	LS: 20% of every extrusion run FAT: 10% of Shipping drums	H	(W)
5)	Hot set test for XLPE insulation	IEC 60811-507 IEC 62067: 2022 C 10.9	Aging Oven Recorder	Elongation under Load: Max. 175 %, Permanent Elongation after Cooling: Max. 15 %	LS: 10% of every extrusion run FAT: 10% of Shipping drums	H	(W)
6)	Insulation shrinkage test	IEC 60840:2020 C 12.5.16	Oven	Max. 4.5 %	LS: 20% of every extrusion run FAT: 10% of Shipping drums	H	(W)
7)	Measurement of resistivity of semiconducting screens	IEC 62067: 2022 C 12.4.11	Oven, Resistivity tester	Conductor Screen : Max. 1,000 Ω.m Insulation Screen : Max. 500 Ω.m	LS: 10% of every extrusion run FAT: 10% of Shipping drums	H	(W)
8)	Measurement of screen protrusions	KM Spec. V3 S3.3.12.2 C1.4.4.8.10	Hot oil chamber	Shall be cylindrical, smooth and free from protrusions and irregularities which extend more than 0.03mm into the insulation	LS: 20% of every extrusion run FAT: 10% of Shipping drums	H	(W)
9)	Measurement of capacitance	IEC 62067: 2022 C 10.10	Capacitance Meter	Max 0.2927㎌/km	LS: 10% of every extrusion run FAT: 10% of Shipping drums	H	(W)

NO	Test Items	Reference Standard /Document	Equipment	Acceptance Criteria	Sampling	Inspection Involvement	
						LS	KM
10)	Measurement of crosslinking by product concentration in XLPE cables	KM Spec. V3 S3.3.12.2 C1.4.4.8.12	TGA Analyzer	a. Total weight change of test sample after 30min : Max. 1.25% of the original sample weight b. Rate of change in the first 5min : Max.0.14 % c. Average rate of change of sample weight between the Time of 15 – 30min.: Max. 0.01%	LS: 20% of every extrusion run FAT: 10% of Shipping drums	H	(W)
11)	FTIR	KM Spec. V3 S3.3.12.2 C1.4.4.8.13	FTIR Analyzer	Comply with the Results of Type Test.	LS: 20% of every extrusion run FAT: 10% of Shipping drums	H	(W)
12)	DSC	KM Spec. V3 S3.3.12.2 C1.4.4.8.13	DSC Analyzer	Comply with the Results of Type Test.	LS: 20% of every extrusion run FAT: 10% of Shipping drums	H	(W)
13)	TGA	KM Spec. V3 S3.3.12.2 C1.4.4.8.13	TGA Analyzer	Comply with the Results of Type Test.	LS: 20% of every extrusion run FAT: 10% of Shipping drums	H	(W)
14)	Cleanliness measurement on the incoming insulation material	KM Spec. V3 S3.3.12.2 C1.4.4.8.13	-	90 ~ 110μm / 100g: Max. 2 EA Bigger than 100μm / 100g: Max. 1 EA	On every batch of incoming material	R	R
15)	Smoothness measurement on the incoming semi-conducting material	KM Spec. V3 S3.3.12.2 C1.4.4.8.13	-	0.05 ~ 0.10mm / 200cm2 : Max. 10 EA 0.10 ~ 0.25mm / 200cm2 : Max. 5 EA Bigger than 0.26mm /200 cm2 : 0 EA	On every batch of incoming material	R	R
16)	Measurement of weight for copper and lead of cable construction	KM Spec. V3 S3.3.12.2 C1.4.4.8	ELECTRIC BALANCE	Copper: Approx 24.6kg/m Lead: Approx 12.0kg/m	LS: 10% of every extrusion run FAT: 10% of Shipping drums	H	(W)
17)	Measurement of density of HDPE & carbon content	KM Spec. V3 S3.3.12.2 C1.4.4.8	Density meter	Density : Min 0.94 g/cm³ Carbon content : 2.5±0.5%	LS: 10% of every extrusion run FAT: 10% of Shipping drums	H	(W)
18)	Water penetration test	IEC 62067:2022 C12.5.14, Annex E, Annex F	Temperature recorder	No Water shall emerge from the ends of the test piece	One sample each 10 km of production length	H	(W)

2.1.3. Sample Test (220kV Power cable)

Table 2-3 Sample Test

NO	Test Items	Reference Standard /Document	Equipment	Acceptance Criteria	Sampling	Inspection Involvement	
						LS	KM
1)	Power frequency voltage test of 5 U0 for 1 Hour	KM Spec. V3 S3.3.12.2 C1.4.4.8.14	Voltage tester	No breakdown occurs at A.C 650kV for 1 hour	One 20m sample from every 10 km of manufacturing length	H	(W)

2.1.4. Routine Test (Joint)

Table 2-4 Routine Test(Joint)

NO	Test Items	Reference Standard /Document	Equipment	Acceptance Criteria	Sampling	Inspection Involvement	
						LS	KM
1)	HV withstand test at 2.5U0 for 30 minutes	KM Spec. V3 S3.3.12.2 C1.4.5.1	AC 600kV Voltage tester	No breakdown at 325kV for 30min	100% (LS) 10% sample (Witness)	H	(W)
2)	PD measurement at 1.5 U0	KM Spec. V3 S3.3.12.2 C1.4.5.1	AC 600kV Voltage tester PD Calibrator	Max. 5pC at 195kV (Background noise Max 2pC)		H	(W)
3)	X-Ray test	KM Spec. V3 S3.3.12.2 C1.4.5.1	X-Ray	Any Crack or Crack Development Shall Not Be Found.		H	(W)
4)	Packing and Preservation Inspection	KM Spec. V3 S3.3.12.2 C1.5	-	The quantities of the products shall be satisfied with packing list and Preservation requirements	100% accessory (LS) 1 sample (Witness)	H	(W)

2.1.5. Sample Test (Joint)

Table 2-5 Sample Test(Joint)

NO	Test Items	Reference Standard /Document	Equipment	Acceptance Criteria	Sampling	Inspection Involvement	
						LS	KM
1)	FTIR	KM Spec. V3 S3.3.12.2 C1.4.5.1	FTIR Analyzer	Comply with the Results of Type Tests	10% (LS) 10% sample (Witness)	H	(W)
2)	Cleanliness measurement on the incoming insulation material	KM Spec. V3 S3.3.12.2 C1.4.5.1	-	90 ~ 110μm / 100g: Max. 2 EA Bigger than 100μm / 100g: Max. 1 EA	On every batch of incoming material	R	R
3)	DLA measurement of insulation at 90℃	KM Spec. V3 S3.3.12.2 C1.4.5.1	DLA Analyzer	Comply with the Results of Type Tests	10% (LS) 10% sample (Witness)	H	(W)
4)	TGA	KM Spec. V3 S3.3.12.2 C1.4.5.1	TGA Analyzer	Comply with the Results of Type Tests	10% (LS) 10% sample (Witness)	H	(W)

2.1.6. Additional Test (Joint)

Table 2-6 Additional Test(Joint)

NO	Test Items	Reference Standard /Document	Equipment	Acceptance Criteria	Sampling	Inspection Involvement	
						LS	KM
1)	Mechanical stretch of each rubber component	KM Spec. V3 S3.3.12.2 C1.4.5.1.1	-	No damage	100% (LS) 10% sample (Witness)	H	(W)
2)	Resistance of outer semi-conducting screen	KM Spec. V3 S3.3.12.2 C1.4.5.1.1	Ohm meter	Max 20kΩ		R	R
3)	Integrity of insulated gap 25kV dc for 15 minutes	KM Spec. V3 S3.3.12.2 C1.4.5.1.1	DC Voltage tester	Shall withstand DC 25kV 15min		H	(W)

2.1.7. Routine Test (Termination)

Table 2-7 Routine Test(Termination)

NO	Test Items	Reference Standard /Document	Equipment	Acceptance Criteria	Sampling	Inspection Involvement	
						LS	KM
1)	HV withstand test at 2.5U0 for 30 minutes	KM Spec. V3 S3.3.12.2 C1.4.5.1	AC 600kV Voltage tester	No breakdown at 325kV for 30min	100% (LS) 10% sample (Witness)	H	(W)
2)	PD measurement at 1.5 U0	KM Spec. V3 S3.3.12.2 C1.4.5.1	AC 600kV Voltage tester PD Calibrator	Max. 5pC at 195kV (Background noise Max 2pC)		H	(W)
3)	X-Ray test	KM Spec. V3 S3.3.12.2 C1.4.5.1	X-Ray	Any Crack or Crack Development Shall Not Be Found.		H	(W)
4)	Packing and Preservation Inspection	KM Spec. V3 S3.3.12.2 C1.5	-	The quantities of the products shall be satisfied with packing list and Preservation requirements	100% accessory (LS) 1 sample (Witness)	H	(W)

2.1.8. Sample Test (Termination)

Table 2-8 Sample Test(Termination)

NO	Test Items	Reference Standard /Document	Equipment	Acceptance Criteria	Sampling	Inspection Involvement	
						LS	KM
1)	FTIR	KM Spec. V3 S3.3.12.2 C1.4.5.1	FTIR Analyzer	Comply with the Results of Type Tests	10% (LS) 10% sample (Witness)	H	(W)
2)	Cleanliness measurement on the incoming insulation material	KM Spec. V3 S3.3.12.2 C1.4.5.1	-	90 ~ 110μm / 100g: Max. 2 EA Bigger than 100μm / 100g: Max. 1 EA	On every batch of incoming material	R	R
3)	DLA measurement of insulation at 90℃	KM Spec. V3 S3.3.12.2 C1.4.5.1	DLA Analyzer	Comply with the Results of Type Tests	10% (LS) 10% sample (Witness)	H	(W)
4)	TGA	KM Spec. V3 S3.3.12.2 C1.4.5.1	TGA Analyzer	Comply with the Results of Type Tests	10% (LS) 10% sample (Witness)	H	(W)

2.1.9. Routine Test (Link Box and Sheath Voltage Limiters)

Table 2-9 Routine Test(Link Box and Sheath Voltage Limiters)

NO	Test Items	Reference Standard /Document	Equipment	Acceptance Criteria	Sampling	Inspection Involvement	
						LS	KM
Link Box							
1)	Visual and dimensional inspection	KM Spec. V3 S3.3.12.6 C1.9.2	Measuring tape	Shall be meet with approved drawings	100% (LS) 10% sample (Witness)	H	(W)
2)	Insulation Resistance measurement test	KM Spec. V3 S3.3.12.6 C1.9.2 b	Insulation resistance measurement	Min. 10MΩ		H	(W)
3)	Contact resistance measurement test	KM Spec. V3 S3.3.12.6 C1.9.2 c	Contact resistance tester	Max. 10μΩ		H	(W)
4)	Water sealing test	KM Spec. V3 S3.3.12.6 C1.9.2 d	Water tank	No leakage of the air from the sample during the test.(0.3bar 15min)		H	(W)
5)	DC withstand test	KM Spec. V3 S3.3.12.6 C1.9.3 ii	DC Voltage tester	No flash over at DC 25kV 5mins		H	(W)
6)	Impulse voltage withstand test	KM Spec. V3 S3.3.12.6 C1.9.3 i	Impulse tester	No flash over ▪ Between links: ± 50kVp / 10 times. ▪ Between links and earth: ± 20kVp / 10 times		H	(W)
7)	Packing inspection	KM Spec. V3 S3.3.12.6 C1.10	-	The quantities of the products shall be satisfied with packing list and Preservation requirements	100% (LS) 1 sample per each type (Witness)	H	(W)
Sheath Voltage Limiters							
1)	Leakage Current Test at DC 5kV for 1 min.	KM Spec V3 S3.3.12.6 C1.9.1	DC Voltage Tester	Leakage Current – Max. 0.1mA	100% (LS) 10% of SVL (Witness)	H	(W)
2)	Insulation Resistance	KM Spec V3 S3.3.12.6 C1.9.1	Insulation resistance measurement	Min. 10MΩ		H	(W)